

EE5507 Project 1 Report

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100

No.	Specification	Value	Achieved	Unit / Condition	
1	DC Gain	> 70	78	dB	10
2	GBW	> 15	30	MHz ($C_{Load} = 1$ pF)	+
3	Phase Margin	> 65	67	Degree	15
4	Input Offset Voltage	< 5	1.43	mV	
5	Input Common Mode Range	0.1 – 1.2	0 – 1.2	V	
6	Output Common Mode Range	0.3 – 1	0.21 – 1	V	

Table 1: OTA Design Specifications and Achieved Performance

1. Refer to “Offset Voltage Simulation gpdk045.pdf”, perform monte-carlo simulation to obtain the offset voltage plot. Estimate the resulting offset voltage. (20%)
2. Include schematic print-out in your report pdf with transistor sizing of your OTA. (10%)
3. Include schematic print-out in your report pdf with DC operating point of your OTA, showing gm, ro, vdsat, vds for all transistors in the schematic using annotation approach. Make sure your figure is big and clear to show the label. (10%)
4. Create a table shown above with one additional column beside the “Value” column with the column name as “Achieved”, and stated your actual achieve performance parameters based on your simulation. For each specification listed in the table, you should include the corresponding simulation results to prove that you have met the specification. (10%)
5. For example, “DC Gain” can be shown with AC analysis. “GBW” and “Phase Margin” can be shown with stability analysis assuming unity gain buffer configuration. “Input Offset Voltage” can be illustrated with Monte Carlo simulation. “Input Common Mode Range” and “Output Common Mode Range” results will be supported by point (3). (30%)
6. Assume voltage supply of 1.2V and indicate clearly your total current consumption based on your simulation results. (You can probe the DC current from the VDD source, 5%)
7. When all the specifications are met, your design will be benchmarked with others according to power consumption, complexity, and how far you exceed the given specifications. (15%)

2 Achieved Performance

[Figure 21](#) and [Figure 22](#) show the sizing and DC operating points of the amplifier, with stages labelled to ease discussion henceforth (Note: these are high resolution, should have no issue zooming in). [Figure 2](#) and [Figure 3](#) show the sizing and DC operating points of the bias generator. [Figure 4](#) and [Figure 5](#) show the sizing and DC operating points of the minimum current selector. [Figure 6](#) shows the top-level schematic measuring the open-loop and closed-loop response. [Figure 7](#) shows the top-level schematic measuring the input offset voltage. The same schematic with DC operating

1

points, [Figure 8](#), shows the total current draw to be 2.8 mA.

2.1 DC Gain & GBW

The differential mode open-loop gain under different input CM levels is plotted against frequency

- 2 marks deducted is from this
- current consumption should be plotted with probing, not read off the DC op point

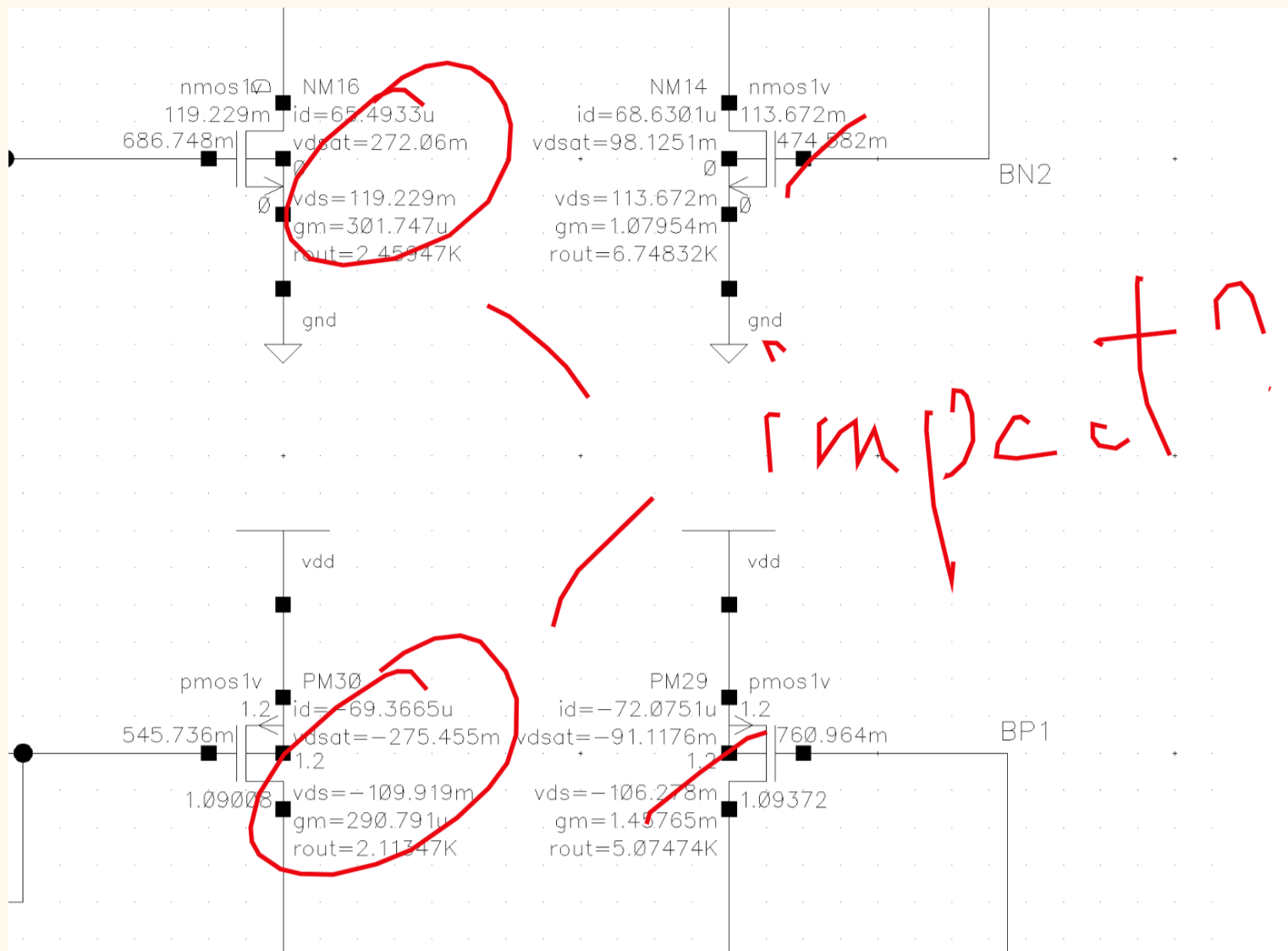
Any gain result
to support?

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Bias Generation



Figure 3: Final Schematic of Bias Generation (DC Operating Point at Input CM = 0.6 V)



- huh???
- for this kind of bias generating circuit, this MOS is expected to be in linear no?

High Swing Cascode 4	Strategy: M6 is designed with a sizing that will make it operate in the linear region $\Rightarrow V_{DS6}$ depends on I_{D6} and V_{GS6} . I_{D6} and V_{GS6} are fixed by I_{ref} and V_{GS7} , respectively. Then V_{DS6} is designed by sizing of M6 to make it $\propto V_{OD1}$
	$V_{GS6} = V_{GS7} \quad V_{DS6} \approx \alpha V_{DSSAT1}$
$V_b = V_{GS5} + V_{DS6}$	※ VGS5 is tracking VGS3 ※ VDS6 is tracking VDS1 with a design margin of $\alpha > 1$
$\left(\frac{W}{L}\right)_{M5} = \left(\frac{W}{L}\right)_{M3} = \left(\frac{W}{L}\right)_{M1}$	$\left(\frac{W}{L}\right)_1 = \left(\frac{\alpha}{1-\sqrt{1-n}}\right)^2 \left(\frac{W}{L}\right)_7$
$\left(\frac{W}{L}\right)_{M7} = n \left(\frac{W}{L}\right)_{M6}$	※ During the analysis no V_{TH} approximation is taken $\Rightarrow$ insensitive to body bias ※ This architecture tracks and eliminate the V_{TH}/body bias sensitivity, but need 3 current source $\Rightarrow$ higher power
M₆ in linear region	※ The device ratio is also fairly large, for $\alpha = 1.1$, $n = 0.5$ and $M1:M7$ is 14.3, then $M2:M1$ is $m \approx 10$. Overall ratio ≈ 100's
M6 is not in saturation by contradiction. $n \leq 1$	※ Note that the device ratio required by this method is fractional, so may not get nice integer multiplier unless nice fractions are chosen specifically (like with n and α)

- or is he talking about the VOD is too big, could push for swing more?



Heng Chun Huat



To: 🌐 Chai Zi Yang

Tue 21-Apr-26 10:46 AM

I see that the vdsat>vds, so I am wondering what is the impact on the overall system. But given this is biasing network, it should be fine. I did not deduct any mark from this.

Best regards,
CH